Lecture Notes

Lecture 1

Vector Spaces and Subspaces

You already know how to add vectors in $\mathbb{R}^3$, multiply matrices, and solve linear systems. This course rebuilds that material on a more abstract foundation, and the first step is the biggest one. Instead of asking *what* a vector is—an arrow? a column of numbers?—we will declare that a vector is **anything that obeys the rules of vector addition and scalar multiplication.** Nothing else about its nature will matter.

This sounds like we are throwing away information, but it is exactly the opposite. By proving theorems from the rules alone, every result we obtain applies *simultaneously* to arrows in space, to matrices, to polynomials, to functions, and—this is why a physicist should care—to the amplitude and state-vector spaces used in quantum mechanics. These are complex vector spaces, so linear combinations of state vectors express superposition. A normalized vector represents a pure physical state only up to phase; mixed states need density operators later in the course. Today's job is to make the linear structure precise: we define fields, vector spaces, and subspaces, and we learn how to build new spaces out of old ones by taking sums and direct sums.

Remark 1.1 (Standing scope for Lectures 1–6). Unless a theorem or displayed example explicitly says otherwise, the algebraic core uses finite-dimensional vector spaces over $\mathbb{R}$ or $\mathbb{C}$. Infinite-dimensional function, polynomial, or sequence examples are labelled as examples or bridges, and do not silently broaden finite-dimension formulas.

Learning objectives

By the end of this lecture you will be able to:

- State the vector-space axioms and check (or refute) them for concrete sets—tuples, matrices, polynomials, functions.
- Explain why the scalar field ($\mathbb{R}$ versus $\mathbb{C}$) is part of the data of a space.
- Apply the three-part subspace test to decide whether a subset is a subspace.
- Form sums and direct sums, and use the zero-test (or $U \cap W = \{0\}$) to decide directness.

1 Scalars: the field

1.1 Fields

Before we can *scale* a vector we must say what numbers we are allowed to scale by. Those numbers form a *field*.

Definition 1.2 (Field). A *field* $\mathbb{F}$ is a set with two operations, addition and multiplication, that are both commutative and associative and are linked by the distributive law $a(b+c) = ab+ac$. Each operation has an identity element (0 and 1, with $1 \neq 0$), every element has an additive inverse, and every *nonzero* element has a multiplicative inverse—so we may divide by any nonzero scalar.

The decisive clause is the last one: in a field you can divide by any nonzero scalar. That single ability is what powers nearly every computation in linear algebra. We will only ever use two fields,

$$\mathbb{R} \quad \text{(the real numbers)} \qquad \text{and} \qquad \mathbb{C} \quad \text{(the complex numbers)},$$

and when a statement holds for both we write $\mathbb{F}$ and call its elements *scalars*.

Remark 1.3. The integers $\mathbb{Z}$ are *not* a field: you cannot divide 1 by 2 and stay in $\mathbb{Z}$. So although $\mathbb{Z}$ has perfectly good addition and multiplication, "a vector space over $\mathbb{Z}$" is not something we allow—we would lose the ability to rescale freely.

1.2 Complex numbers, briefly

Because the standard quantum formalism uses complex amplitudes, we pause to recall the arithmetic you will lean on all semester. A complex number is written $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$; its *real* and *imaginary* parts are a and b, its *conjugate* is $\overline{z} = a - bi$, and its *modulus* is

$$|z| = \sqrt{z\overline{z}} = \sqrt{a^2 + b^2}.$$

Division is carried out by "rationalising" with the conjugate, e.g.

$$\frac{1}{2+i} = \frac{2-i}{(2+i)(2-i)} = \frac{2-i}{5}.$$

Every nonzero complex number also has a *polar form*

$$z = r\,e^{i\theta} = r(\cos\theta + i\sin\theta), \qquad r = |z| > 0, \tag{1.1}$$

where the argument θ is determined modulo 2π. The number 0 has no argument and is handled separately. For nonzero factors multiplication is then strikingly simple: moduli multiply and arguments add modulo 2π, $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2\, e^{i(\theta_1+\theta_2)}$.

Example 1.4 (Polar form in action). Write $1 + i$ in polar form and compute $(1 + i)^4$. Here $r = \sqrt{1^2 + 1^2} = \sqrt{2}$ and $\theta = \frac{\pi}{4}$, so $1 + i = \sqrt{2}\, e^{i\pi/4}$. Raising to the fourth power multiplies the argument by 4 and the modulus to the fourth:

$$(1+i)^4 = (\sqrt{2})^4 e^{i\pi} = 4\,(\cos\pi + i\sin\pi) = -4.$$

The polar form turned a messy binomial expansion into one line. ◁

Remark 1.5. The factor $e^{i\theta}$, a complex number of modulus 1, is the mathematical home of a quantum-mechanical *phase*. Interference—the phenomenon at the heart of the double-slit experiment—is the addition of such phases, which is why the standard formulation uses a complex space. A real formulation needs an additional compatible complex structure to retain this phase multiplication.

2 Vector spaces

Definition 1.6 (Vector space). A *vector space over* $\mathbb{F}$ is a set V together with an *addition* $(u, v) \mapsto u + v \in V$ and a *scalar multiplication* $(a, v) \mapsto av \in V$ (for $a \in \mathbb{F}$ and $u, v \in V$) satisfying, for all $u, v, w \in V$ and $a, b \in \mathbb{F}$:

1. $u + v = v + u$ (commutativity);
2. $(u + v) + w = u + (v + w)$ and $(ab)v = a(bv)$ (associativity);
3. there exists $0 \in V$ with $v + 0 = v$ (additive identity);
4. for each v there is $w \in V$ with $v + w = 0$ (additive inverse);
5. $1\,v = v$ (multiplicative identity);
6. $a(u + v) = au + av$ and $(a + b)v = av + bv$ (distributivity).

Elements of V are *vectors*. A vector space over $\mathbb{R}$ is a *real* vector space; over $\mathbb{C}$, a *complex* vector space. We abbreviate $u + (-v)$ as $u - v$.

Remark 1.7. Notice what the definition does *not* say: it never specifies what a vector is. A vector might be a tuple of numbers, a matrix, a polynomial, a function, or a quantum amplitude/state vector. All that is required is that the eight rules hold. Prove a theorem from the rules and you have proved it everywhere at once—that is the whole strategy of the subject.

The axioms resemble the ordinary rules of arithmetic, so several "obvious" facts must now be *derived* rather than assumed. The proofs are short but worth following closely: they are your first taste of reasoning from axioms.

Proposition 1.8 (Elementary consequences). Let V be a vector space over $\mathbb{F}$. Then

1. the additive identity 0 is unique, and each $v \in V$ has a unique additive inverse, written $-v$;
2. $0\,v = 0$ for every $v \in V$ (the left 0 is the scalar, the right 0 the zero vector);
3. $a\,0 = 0$ for every $a \in \mathbb{F}$;
4. $(-1)v = -v$ for every $v \in V$.

Proof. (1) If 0 and $0'$ are both additive identities then $0' = 0' + 0 = 0 + 0' = 0$, using axiom (3) for each and commutativity. For inverses, if $v + w = 0$ and $v + w' = 0$ then

$$w = w + 0 = w + (v + w') = (w + v) + w' = 0 + w' = w'.$$

(2) By distributivity, $0\,v = (0 + 0)v = 0\,v + 0\,v$. Adding the inverse of $0\,v$ to both sides gives $0 = 0\,v$.

(3) Similarly $a\,0 = a(0 + 0) = a\,0 + a\,0$, so $a\,0 = 0$.

(4) We show $(-1)v$ is *an* additive inverse of v; uniqueness from (1) then identifies it with $-v$. Indeed, using axiom (5), distributivity and (2),

$$v + (-1)v = 1\,v + (-1)v = \big(1 + (-1)\big)v = 0\,v = 0. \qquad \blacksquare$$

With addition and scaling in hand we can name the central construction of the whole subject.

Definition 1.9 (Linear combination). A *linear combination* of vectors $v_1, \dots, v_m \in V$ is any vector of the form $a_1 v_1 + \cdots + a_m v_m$ with scalars $a_1, \dots, a_m \in \mathbb{F}$.

Almost everything ahead is phrased in this language. In physics terms, the superposition principle says that linear combinations live in the ambient amplitude/state-vector space. A nonzero combination must still be normalized to represent a pure state, and vectors differing by an overall phase represent the same pure-state ray. Which vectors a given set can reach by linear combination—its *span*—is the systematic subject of the next lecture.

3 A gallery of examples

The reward for abstraction is the sheer variety of objects the definition captures. In every example below, addition and scalar multiplication are performed *entrywise* (or pointwise), and the axioms then hold automatically because they already hold for the scalars—so we get the verification "for free."

Example 1.10 (Coordinate space $\mathbb{F}^n$). $\mathbb{F}^n$ is the set of n-tuples $(x_1, \dots, x_n)$ with $x_k \in \mathbb{F}$, added and scaled componentwise. The zero vector is $(0, \dots, 0)$. Taking $\mathbb{F} = \mathbb{R}$ and $n = 3$ recovers the familiar $\mathbb{R}^3$; taking $\mathbb{F} = \mathbb{C}$ gives the amplitude/state-vector space for a finite n-level quantum model (for $n = 2$, a qubit). Normalized vectors represent pure-state rays rather than distinct physical states. ◁

Example 1.11 (Matrices and polynomials). $M_{m\times n}(\mathbb{F})$, the $m \times n$ matrices over $\mathbb{F}$, is a vector space under entrywise operations, with the zero matrix as zero vector. So is $\mathcal{P}(\mathbb{F})$, the set of all polynomials $p(t) = a_0 + a_1 t + \cdots + a_d t^d$ with coefficients in $\mathbb{F}$, and so is its subset $\mathcal{P}_m(\mathbb{F})$ of polynomials of degree *at most* m. The words "at most" matter: the polynomials of degree *exactly* m are *not* a vector space, because $t^m + (-t^m)$ has smaller degree. ◁

Example 1.12 (Functions and sequences). For a nonempty set S, let $\mathbb{F}^S$ be the set of functions $f\colon S \to \mathbb{F}$, with $(f+g)(x) = f(x)+g(x)$ and $(af)(x) = a\,f(x)$. This is a vector space whose zero vector is the function that is 0 everywhere. For instance, in $\mathbb{R}^\mathbb{R}$ with $f = \sin$ and $g = \cos$, the vector $2f - 3g$ is the function $x \mapsto 2\sin x - 3\cos x$, taking the value -3 at $x = 0$. Two special cases recur constantly:

- $S = \{1, 2, 3, \dots\}$ gives the space $\mathbb{F}^\infty$ of infinite *sequences* $(x_1, x_2, \dots)$;
- $S = [a, b]$ gives function spaces on an interval, including the elementary real example $C[a, b]$. Standard wavefunctions instead use complex square-integrable amplitudes, introduced with L^2 in Lecture 13; they need not be continuous. ◁

Example 1.13 (The same set, two different spaces). The set $\mathbb{C}$ can be regarded as a *complex* vector space (scalars from $\mathbb{C}$) *or* as a *real* vector space (scalars from $\mathbb{R}$ only). These are genuinely different structures: over $\mathbb{C}$ every element is a scalar multiple of 1, whereas over $\mathbb{R}$ we need both 1 and i to reach every point. More generally $\mathbb{C}^n$ may be viewed as a real vector space, and doing so doubles the number of independent directions. The field you scale by is part of the data of a vector space, never an afterthought—a point that returns the moment we count dimensions next lecture. ◁

Example 1.14 (State spaces in physics). Two examples tie directly to what you know and to what is coming.

1. *Homogeneous linear systems.* The solution set $\{x \in \mathbb{F}^n : Ax = 0\}$ is a vector space: if $Ax = 0$ and $Ay = 0$ then $A(ax + by) = aAx + bAy = 0$. You already met this set as the null space when row-reducing A.
2. *The harmonic oscillator.* Fix $\omega > 0$. The twice-differentiable solutions $y\colon \mathbb{R} \to \mathbb{R}$ of $\ddot{y} + \omega^2 y = 0$ form a real vector space, because if y_1, y_2 are solutions then so is $ay_1 + by_2$ (differentiation is linear). Explicitly the solution space is $\{A\cos\omega t + B\sin\omega t : A, B \in \mathbb{R}\}$—we will see next lecture that it is two-dimensional. Closure under linear combinations is precisely the *superposition principle*. ◁

Example 1.15 (A non-example). Not everything is a vector space. The line $L = \{(x, y) \in \mathbb{R}^2 : y = x + 1\}$ (Figure 1) is not one: it misses the origin, so it has no zero vector, and it is not closed—adding the points $(0, 1)$ and $(1, 2)$ of L gives $(1, 3)$, which is off the line. We will see in Section 4 that "passes through the origin" is exactly the repair. ◁

4 Subspaces

Many vector spaces arise sitting *inside* a larger one: a line through the origin inside $\mathbb{R}^3$, the continuous functions inside all functions.

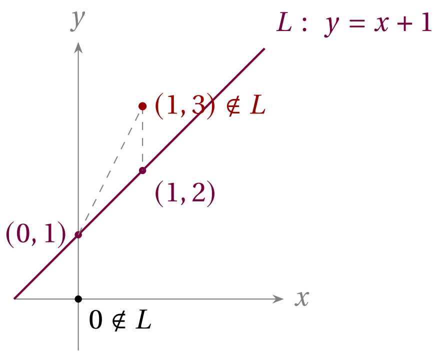

Figure 1: A line not through the origin fails both the zero-vector and closure conditions. The sum of two points on L leaves L.

Definition 1.16 (Subspace). A subset $U \subseteq V$ is a *subspace* of V if U is itself a vector space under the addition and scalar multiplication inherited from V.

Checking all eight axioms each time would be tedious—and unnecessary. Almost every axiom is inherited from V for free; only three things can go wrong, and the next theorem isolates them.

Theorem 1.17 (Subspace test). A subset $U \subseteq V$ is a subspace of V if and only if

1. $0 \in U$;
2. $u + w \in U$ whenever $u, w \in U$ (closed under addition);
3. $au \in U$ whenever $a \in \mathbb{F}$ and $u \in U$ (closed under scaling).

Proof. If U is a subspace it is a vector space, hence closed under both operations and in possession of a zero (which must coincide with the 0 of V). Conversely, assume (1)–(3). Conditions (2) and (3) guarantee the two operations map U into U, and (1) supplies the additive identity. Each $u \in U$ has its inverse inside U as well, since $-u = (-1)u \in U$ by (3) and Proposition 1.8(4). The remaining axioms (commutativity, associativity, distributivity, $1u = u$) are identities that already hold for all of V, so in particular they hold on U. Hence U is a vector space, i.e. a subspace. ■

Remark 1.18. Condition (1) may be replaced by "U is nonempty": pick any $u \in U$, then $0\,u = 0 \in U$ by closure. In practice the quickest disqualifier runs the other way—if $0 \notin U$, stop, it is not a subspace.

Example 1.19 (Each test condition is necessary). None of the three conditions is redundant. Here are three subsets of $\mathbb{R}^2$, each failing exactly one of them.

- The integer lattice $\mathbb{Z}^2 = \{(m, n) : m, n \in \mathbb{Z}\}$ contains 0 and is closed under addition, but not under scaling, since $\frac{1}{2}(1, 0) = (\frac{1}{2}, 0) \notin \mathbb{Z}^2$.
- The pair of quadrants $\{(x, y) : xy \geq 0\}$ contains 0 and is closed under scaling, but not under addition: $(1, 0)$ and $(0, -1)$ both lie in it, yet their sum $(1, -1)$ has $xy = -1 < 0$.
- The empty set $\varnothing$ is vacuously closed under addition and scaling (there are no vectors to test), but it does not contain 0.

A genuine subspace must pass all three checks. ◁

Example 1.20 (Subspaces and non-subspaces). Inside $\mathbb{R}^3$, every line and plane *through the origin* is a subspace; a line not through the origin is not. Inside $M_n(\mathbb{F})$, each of the following is a subspace, because its defining condition survives addition and scaling: the symmetric matrices ($A^\top = A$), the antisymmetric matrices ($A^\top = -A$), the upper-triangular matrices, and the trace-zero matrices ($\operatorname{tr} A = 0$). By contrast for $n \geq 2$ the singular matrices $\{A : \det A = 0\}$ are *not* a subspace: $\operatorname{diag}(1, 0, \dots, 0)$ and $\operatorname{diag}(0, 1, \dots, 1)$ are singular, yet their sum is I_n. For $n = 1$ the singular set is just $\{0\}$, which is a subspace. ◁

Example 1.21 (A subspace defined by a linear condition). Is $U = \{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$ a subspace of $\mathbb{F}^3$? Run the test. First $0 = (0, 0, 0)$ satisfies $0 = 5 \cdot 0$, so $0 \in U$. Next, if $x_1 = 5x_3$ and $y_1 = 5y_3$ then $(x_1 + y_1) = 5(x_3 + y_3)$, so $x + y \in U$; and $(ax_1) = 5(ax_3)$, so $ax \in U$. All three conditions hold, so U is a subspace. What made it work is that the defining equation is *linear and homogeneous*—contrast the non-example $y = x + 1$ of Example 1.15, whose constant term destroyed closure. ◁

5 Sums and direct sums

Given two subspaces $U, W \subseteq V$, how do we combine them into a third? There are two natural attempts—intersection and union—and they behave very differently.

Proposition 1.22 (Intersection is a subspace). If U and W are subspaces of V, then $U \cap W$ is a subspace of V.

Proof. Both U and W contain 0, so $0 \in U \cap W$. If $x, y \in U \cap W$ and $a \in \mathbb{F}$, then $x + y$ and ax lie in U (because $x, y \in U$ and U is a subspace) and likewise in W, hence in $U \cap W$. The three conditions of the subspace test (Theorem 1.17) hold, so $U \cap W$ is a subspace. ■

Example 1.23 (Two planes meet in a line). In $\mathbb{R}^3$ take the planes $U = \{(x, y, z) : z = 0\}$ and $W = \{(x, y, z) : y = 0\}$, both subspaces through the origin. Their intersection $U \cap W = \{(x, 0, 0)\}$ is the x-axis—again a subspace, exactly as Proposition 1.22 promises. ◁

The *union*, by contrast, is almost never a subspace: the union of the two coordinate axes in $\mathbb{R}^2$ contains $(1, 0)$ and $(0, 1)$ but not their sum $(1, 1)$ (Figure 2). To build a new subspace from U and W we therefore use neither operation directly, but the *sum*.

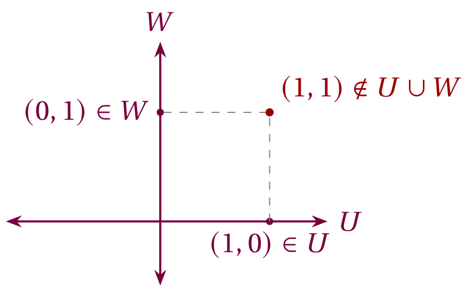

Figure 2: The union of two subspaces need not be a subspace: the sum of a vector on the x-axis and one on the y-axis lies on neither axis.

Definition 1.24 (Sum of subspaces). If $U_1, \dots, U_m$ are subspaces of V, their *sum* is

$$U_1 + \cdots + U_m = \{u_1 + \cdots + u_m : u_k \in U_k\}.$$

A quick check with the subspace test shows $U_1 + \cdots + U_m$ is again a subspace; in fact it is the *smallest* subspace of V that contains every U_k (any subspace containing all the U_k must contain all their sums).

A sum is most useful when each vector decomposes in only *one* way.

Definition 1.25 (Direct sum). The sum $U_1+\cdots+U_m$ is a *direct sum*, written $U_1\oplus\cdots\oplus U_m$, if every vector in it has *exactly one* expression as $u_1+\cdots+u_m$ with $u_k\in U_k$.

To test directness we need not inspect every vector—only the zero vector.

Proposition 1.26 (Zero-test for directness). $U_1+\cdots+U_m$ is direct if and only if the only way to write $0=u_1+\cdots+u_m$ with $u_k\in U_k$ is $u_1=\cdots=u_m=0$.

Proof. If the sum is direct, 0 has a unique decomposition; the all-zero one works, so it is the only one. Conversely, suppose 0 decomposes only trivially and take any v with two decompositions $v=\sum_k u_k=\sum_k u_k'$. Subtracting, $0=\sum_k(u_k-u_k')$ with $u_k-u_k'\in U_k$, so by hypothesis each $u_k-u_k'=0$; the two decompositions coincide. ■

For two subspaces the criterion takes a memorable geometric form.

Corollary 1.27 (Two-subspace criterion). For subspaces $U,W\subseteq V$,

$$U+W \text{ is a direct sum} \iff U\cap W=\{0\}.$$

Proof. ($\Rightarrow$) If $v\in U\cap W$ then $0=v+(-v)$ with $v\in U$ and $-v\in W$; directness forces $v=0$. ($\Leftarrow$) If $U\cap W=\{0\}$ and $0=u+w$ with $u\in U$, $w\in W$, then $u=-w\in U\cap W$, so $u=0$ and hence $w=0$; Proposition 1.26 then gives directness. ■

Example 1.28 (Decomposing $\mathbb{R}^3$). Let $U=\{(x,y,0)\}$ be the xy-plane and $W=\{(0,0,z)\}$ the z-axis. Every vector splits as $(x,y,z)=(x,y,0)+(0,0,z)$ (Figure 3), and $U\cap W=\{0\}$, so $\mathbb{R}^3=U\oplus W$. The splitting is unique—which is just what directness means. ◁

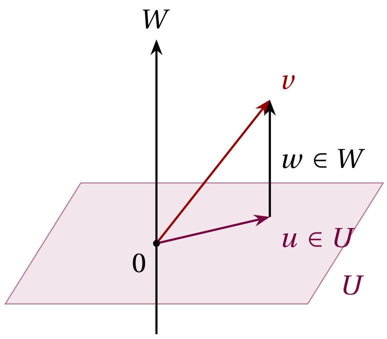

Figure 3: $\mathbb{R}^3=U\oplus W$: each v splits uniquely as a vector u in the plane plus a vector w along the axis.

Example 1.29 (Symmetric and antisymmetric matrices). Let Sym and Skew be the symmetric and antisymmetric subspaces of $M_n(\mathbb{F})$. Any matrix splits as

$$A=\underbrace{\tfrac{1}{2}(A+A^{\mathsf{T}})}_{\in\,\mathrm{Sym}}+\underbrace{\tfrac{1}{2}(A-A^{\mathsf{T}})}_{\in\,\mathrm{Skew}}, \tag{1.2}$$

so $M_n(\mathbb{F})=\mathrm{Sym}+\mathrm{Skew}$; and a matrix that is both symmetric and antisymmetric satisfies $A=A^{\mathsf{T}}=-A$, forcing $A=0$, so the intersection is trivial. By Corollary 1.27, $M_n(\mathbb{F})=\mathrm{Sym}\oplus\mathrm{Skew}$.

For example,

$$\begin{pmatrix} 1 & 2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 1 & 3 \end{pmatrix} + \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

the unique symmetric-plus-antisymmetric decomposition prescribed by (1.2). (This very splitting underlies the decomposition of a physical strain or field gradient into a deformation and a rotation.) ◁

Remark 1.30 (Why this matters in physics). The same algebra returns at the climax of the course. In quantum mechanics an ideal projective measurement uses an orthogonal direct sum of eigenspaces. Uniqueness identifies the components; the later inner product, normalization, spectral projectors, and Born postulate turn their squared norms into probabilities. Today's sums and direct sums are part of the scaffolding for that finite model in Lecture 10.

Summary of main concepts

A vector space is a set closed under addition and scalar multiplication and obeying eight axioms; what the vectors "are" is irrelevant. Coordinate n-tuples, matrices, polynomials, functions, and quantum state-vector spaces are all examples. A subspace is a subset that is closed under both operations and contains 0—check exactly those three things. Subspaces combine through sums, and a sum is *direct* precisely when the zero vector has only the all-zero decomposition. For two subspaces this is equivalent to their intersection being $\{0\}$; pairwise-trivial intersections do not suffice for three or more. The standard quantum examples use $\mathbb{C}$ because complex amplitudes carry phase; a bare real space does not encode that multiplication without an added compatible complex structure.

- **Field**—scalars you can add, multiply, and divide by; here always $\mathbb{R}$ or $\mathbb{C}$.
- **Vector space**—a set closed under addition and scalar multiplication satisfying eight axioms—the nature of the vectors is irrelevant.
- **Linear combination**—$a_1 v_1 + \cdots + a_m v_m$; quantum superposition occurs in the ambient amplitude/state-vector space.
- **Subspace**—a subset containing 0 and closed under both operations (the three-part test).
- **Sum and direct sum**—$U_1 + \cdots + U_m$; *direct* ($\oplus$) when every vector decomposes uniquely, equivalently when 0 does.

Exercises for this lecture are in the companion sheet `exercises-1.pdf`*.*